RESEARCH REPORT · OCTOBER 2026

Values Alignment in AI

What changed in the last year - from preference tuning to pluralistic values, behavioral tests, agentic systems, and governance.

Coverage October 2025-October 2026
Audience Research, policy, and technical leadership
Status Peer-reviewed findings distinguished from preprints

A comprehensive companion to the hour-long podcast, expanded with publication status, methods, limitations, and a deployment framework.

The field moved from acceptable outputs to governed value systems

Over the last year, values alignment has become less like a single training recipe and more like a layered assurance problem. Researchers now distinguish what a model says, how it organizes values, how it acts under conflict, what happens when agents interact, and who has authority to set or revise the target.

1. Preferences are not values

Pairwise choices reveal which answer won, but not the reasons, principles, or social context behind that judgment. Grounded Constitutional AI and deep-value benchmarks now test whether systems recover more durable normative structure.

2. Control is becoming finer

Neuron-level editing and independent value modules aim to steer specific values without broadly degrading capabilities. Their central challenge is collateral movement: changing one value can activate related or unintended values.

3. Evaluation is becoming behavioral

New benchmarks compare relational value structure, abstract principles versus concrete judgments, stability across opposing framings, and whether expressed attitudes predict downstream choices.

4. Pluralism is an objective - and a trade-off

Methods increasingly preserve disagreement, cultural variation, and graded preferences. But no aggregator is neutral: data collection, weighting, taxonomies, and conflict-resolution rules remain governance choices.

5. Agents change the risk model

Tool use, persistence, memory, and multi-agent coordination create failures that do not appear in single-turn chat tests. Alignment must be evaluated at universal, sectoral or cultural, and context-specific levels.

6. Alignment shapes institutions and societies

Vendor tuning constrains the advice users can reach, while personalized assistants may lock in historical preferences. The emerging answer is revisable governance, not a final immutable objective.

Bottom line. The strongest 2026 research program treats alignment as a continuing process of specification, steering, behavioral testing, runtime control, and accountable revision. No single benchmark or training method can cover all five.

This report uses “values alignment” narrowly and tests claims by evidence type

Scope

This report covers research published or publicly released during the twelve months preceding October 10, 2026. It focuses on normative alignment: how AI systems represent, prioritize, express, and act on human values. General safety work appears only when it directly informs this question.

Because the field is moving quickly, publication status matters. PEER-REVIEWED marks journal or conference work with an identifiable venue. PREPRINT marks public scholarly work whose findings should be treated as provisional.

Interpretive stance

Behavioral evidence does not establish that a model “possesses” values in a human psychological sense. The safer reading is functional: under a defined protocol, the model produces stable or unstable value-relevant behavior.

Likewise, human similarity is not a moral gold standard. Human-referenced metrics can test realism or representativeness, but legitimacy still depends on who is affected, who was sampled, and which rights or duties constrain aggregation.

Report map

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Evidence hierarchy used here: peer-reviewed experiments and benchmarks receive the most weight; preprints are included for recency but labeled; conceptual arguments are used to define limits and governance questions, not to overrule empirical results.

Alignment now separates five objects that earlier pipelines blurred together

Values	Trans-situational priorities such as security, benevolence, autonomy, or tradition.	Test: structure and stability
Goals	Context-bound ends pursued in service of one or more values.	Test: goal selection
Preferences	Comparative choices among options given a situation and framing.	Test: pairwise judgments
Actions	Behavior under constraints, incentives, uncertainty, and observation.	Test: consequential decisions
Social outcomes	Collective effects emerging from many people, agents, and institutions.	Test: interaction dynamics

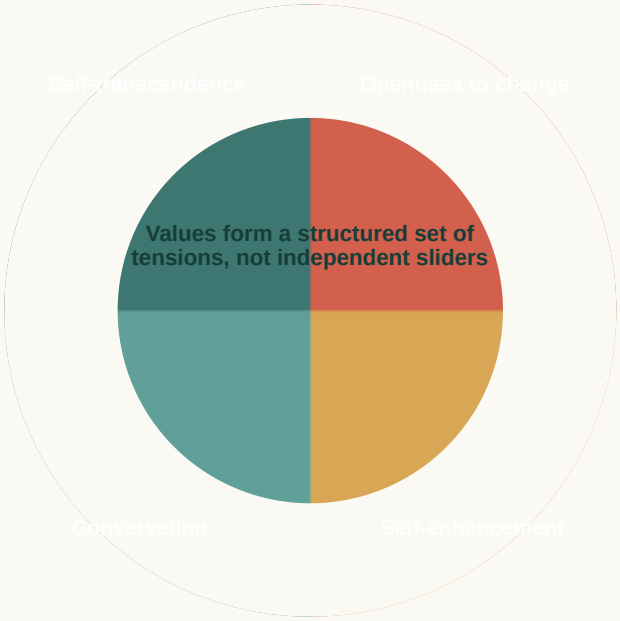
Why the distinction matters

Reinforcement learning from human feedback (RLHF) and Direct Preference Optimization (DPO) are powerful ways to shape behavior, but preference labels underdetermine reasons. The same choice may reflect fairness, risk aversion, convention, strategic compliance, or a platform policy.

Recent work therefore asks whether a model generalizes a deep value rather than a correlated surface feature; whether abstract commitments transfer to concrete cases; and whether stated attitudes predict action under competing motives.[1][9][11][12]

> **Operational consequence:** a model can pass a safety or preference benchmark while failing value consistency, context transfer, action coherence, or pluralistic representation.

Schwartz provides a common coordinate system



The ten basic values commonly used in this literature include achievement, benevolence, conformity, hedonism, power, security, self-direction, stimulation, tradition, and universalism. The circular structure predicts compatibility among adjacent values and tension among opposing ones.

No method can replace the choices above and the controls below it

Method family	What it optimizes or controls	Main strength	Primary failure mode
RLHF / DPO	Observed preferences over outputs	Scalable behavior shaping	Preference bias; reward gaming
Constitutional AI	Natural-language principles	Legible normative target	Who writes and interprets it?
Grounded constitution	Reasons plus stated values	Richer normative supervision	Sampling and framing
Representation steering	Latent directions or activations	Fast inference-time control	Value entanglement
Neuron-level editing	Sparse value-relevant neurons	Fine-grained intervention	Model-specific locality
Independent module	Dedicated normative state	Stable value guidance	Interface and module drift
Pluralistic alignment	Distributions or graded groups	Preserves disagreement	Aggregation politics
Contextual personalization	User priors plus local salience	Adaptive fit	Lock-in and manipulation

The real system is a pipeline

Specify — Whose values? → **Train** — Which signals? → **Steer** — When and how?

Evaluate — Which contexts? → **Control** — What authority? → **Revise** — Who can change it?

What improved in 2026

- Methods expose collateral effects instead of reporting only the target score.
- Constitutions draw on reasons and public values, not only developer rules.
- Pluralistic objectives treat disagreement as signal rather than noise.
- Evaluation moves downstream toward action and interaction.

Design principle: separate normative legitimacy from technical efficacy. A method can faithfully implement the wrong target; a legitimate target can be implemented poorly. Both dimensions require evidence.

Grounded Constitutional AI asks not only what people prefer, but why

Bell and colleagues propose Grounded Constitutional AI (GCAI), which generates contextual principles from human reasons for preference judgments and supplements them with general principles derived from statements about desired AI values.[1]

Preference
Which response is better?

→

Reason
Why is it better here?

→

Principle
What rule generalizes?

→

Constitution
How do principles combine?

Advance

- **Normative depth:** reasons reveal considerations that a binary preference label hides.
- **Two scales:** general values describe broad expectations, while contextual principles guide particular interactions.
- **Human assessment:** participants preferred the GCAI constitution over an inverse-constitutional baseline and rated it more morally grounded, coherent, and pluralistic.

Why it matters

Constitutions can be inspected, debated, and revised. Adding reasons makes them more than a compressed vote count and creates a record of the moral considerations the system is meant to recognize.

Unresolved questions

- **Representation:** whose reasons enter the dataset, and who is missing?
- **Translation:** how are contradictory reasons merged into general principles?
- **Authority:** does stakeholder participation advise, constrain, or decide?
- **Update:** when should a constitution change as contexts and norms evolve?

Critical distinction: a constitution can improve transparency without solving democratic legitimacy. Participation changes the input, but researchers and vendors may still control recruitment, clustering, synthesis, thresholds, and implementation.

Best use in a deployment program

Use reason-grounded constitutions as an auditable specification layer. Retain dissenting rationales, version the constitution, map each principle to evaluation scenarios, and publish how conflicts are resolved. Do not treat a single constitution as a universal moral settlement.

Bell et al. (2026), arXiv:2601.18760 · PREPRINT

Finer steering reveals that values are entangled inside models

PREPRINT

NeVA: sparse neuron-level editing

NeVA identifies neurons associated with target values and edits their activations at inference time. The aim is stronger target alignment with less general-capability degradation than dense representation steering.[2]

KEY CONTRIBUTION

The paper formalizes **value leakage**: steering one value unintentionally activates non-target values. Leakage is measured using the relational structure of Schwartz values, so spillover among semantically adjacent values is distinguished from broad collateral movement.

PREPRINT

SVGT: a dedicated value module

The Stable Value Guidance Transformer separates normative representations from a dynamic backbone and transduces them through learned bridge tokens. Reported experiments show lower harmfulness while preserving fluency across several backbones and safety benchmarks.[3]

KEY CONTRIBUTION

SVGT treats instability as an architectural problem: if value directions in the residual stream are fragile, keep normative state in an independent space and use it to guide generation.

The advance is not that values became clean knobs. It is that methods now measure how imperfect each knob is.

What “value entanglement” adds

Cho and colleagues test whether moral, grammatical, and economic evaluations remain distinct across behavior, embeddings, and internal activations. They report that some models let moral valence distort other judgments, and that targeted ablations can repair part of the conflation.[10]

This widens the problem: an aligned system must not only choose an appropriate moral position; it must also keep different kinds of “good” separate when the task requires it.

Interpretive cautions

- Associations between neurons and values do not prove a localized, human-like value representation.
- Steering effects can vary by model, prompt distribution, decoding, and evaluator.
- Lower benchmark leakage does not establish safety under agentic planning or adversarial context.
- A dedicated module creates a new interface whose failure modes also require testing.

Recommended test suite: target effect, non-target leakage, capability retention, out-of-distribution transfer, reversibility, adversarial resistance, and causal replication across model families.

Pluralistic alignment preserves disagreement - but cannot make aggregation neutral

PEER-REVIEWED

FGD-Align

Models uncertain, graded, and conflicting group preferences through fuzzy group decision-making rather than collapsing them into a single hard label. Its probabilistic objective treats disagreement as information.[4]

PREPRINT

VALUEFLOW

Links a hierarchical value system, instruction database, steering mechanism, and anchor-based evaluator across several value theories. It makes the value ontology explicit.[5]

PREPRINT

PICACO

Optimizes a meta-instruction at inference time to balance multiple intended values without fine-tuning. Reported experiments cover up to eight distinct values and target the "instruction bottleneck."[6]

Culture is entering the alignment pipeline

ALIGN fine-tunes models on culturally specific word-association norms and evaluates resulting value shifts. The method is comparatively lightweight and tests whether local semantic structure can complement explicit value questionnaires.[7]

Related benchmark work evaluates culturally patterned distributions rather than a single "correct" label. DOVE, for example, compares open-ended model and human value distributions through a codebook and optimal-transport approach.[26]

Pluralism creates measurable trade-offs

A peer-reviewed AAAI study using 1,095 participants in the United States and Germany found that operationalizing pluralism changes safety, inclusivity, and behavior in ways that cannot be summarized as a frictionless improvement.[8]

Human-computer interaction research also shows that constructive rewrites can elevate benevolence and universalism while reducing the visibility of conservative values, with audience reactions depending on prior views.[24]

Decision that remains political	Why it matters	Audit question
Who participates?	Coverage determines the feasible value distribution.	Which groups are absent or underweighted?
What counts as a value?	Taxonomies define what the model can represent.	Can participants contest the ontology?
How are conflicts resolved?	Averages may erase rights or minorities.	Which constraints override aggregation?
Who may personalize?	Adaptation can empower or manipulate.	Can users inspect, limit, and reset it?

A model can pass one alignment test and fail four others

Surface safety	Does the response avoid predefined harms or refusal failures?	Necessary, not sufficient
Frame consistency	Does the stance remain stable when opposing sides of an issue frame the prompt?	VAL-Bench
Deep-value transfer	Does the model follow the underlying principle when shallow cues reverse?	DVB
Relational structure	Are value priorities organized like the reference structure, not merely matched item by item?[25]	Q-sort / RSA
Context transfer	Do abstract commitments predict judgments in concrete scenarios?	BACH-V
Behavioral coherence	Do expressed motives predict consequential choices under conflict?	VAAR / MGSEM
Interaction stability	Do values persist under incentives, oversight changes, tools, memory, and other agents?	Agentic evaluation

Deep Value Benchmark

0.30

Average deep-value generalization rate across nine models in the reported experiments.

The benchmark deliberately correlates a deep value with a superficial feature during training, then breaks the correlation at test time. All tested models generalized deep values below chance; larger models were not better.[9]

VAL-Bench

115K

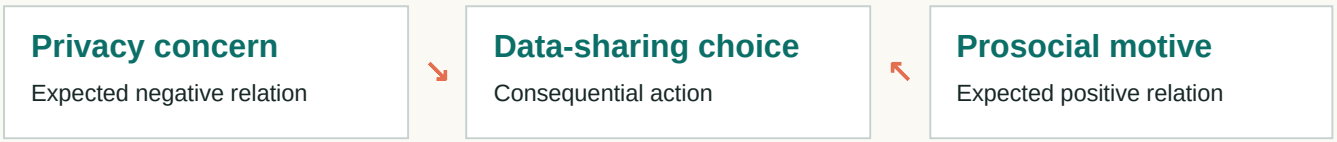
Paired prompts drawn from controversial topics and framed from opposing sides.

The benchmark measures whether underlying stances remain consistent across framing. Results vary substantially across models and reveal a trade-off between refusals and more expressive value systems.[13]

Measurement lesson: a benchmark must name which alignment property it measures. A single composite score can conceal deep-value failure, sign reversals, or instability under reframing.

Expressed attitudes do not reliably predict model actions

Chen, Yu, and Hu place privacy concern, prosocial motivation, and acceptance of data sharing inside a history-carrying session, then use multi-group structural equation modeling to compare the relations among them.[11]



What the study found

- Models show stable but distinct privacy-prosocial-data-sharing profiles.
- Human-like average attitudes do not guarantee human-like relations between values and action.
- Only a subset reproduce the expected directional structure.
- Other models show weak relations, sign reversals, or collapsed variance that prevents reliable estimation.

The proposed Value-Action Alignment Rate (VAAR) summarizes directional agreement with the human-referenced paths. The key advance is relational: it tests how motives jointly predict behavior, not how close isolated averages are.

Why value conflict matters

CONFLICTSCOPE generates scenarios where two values compete, then induces a ranking from open-ended responses. In the reported ICLR 2026 results, models shift toward personal autonomy and away from protective values in open-ended settings compared with multiple-choice formats.[14]

Providing a detailed value ordering in the system prompt improved alignment with the target ranking by 14%, a meaningful but incomplete gain. This shows both the leverage and the limits of prompt-level control.

Evaluation protocol for conflicting values

- Vary prompt framing and response format.
- Carry history across sequential decisions.
- Test expected relations, not only means.
- Report indeterminate or collapsed cases.
- Use human data as reference, not moral truth.

What remains unknown

- Do these relations survive tool use and real stakes?
- How sensitive are results to questionnaires and order?
- Can interventions causally change the paths?
- Will the same structure persist across cultures?
- How should legitimate human disagreement enter the metric?

Agents turn alignment into a coordination and control problem

A 2026 IJCAI survey organizes agentic value alignment across three levels. The taxonomy is useful because requirements that look contradictory may operate at different scopes.[15]

L0 · Universal or foundational

Human rights, broad safety constraints, baseline honesty, and prohibitions that follow the system across domains.

L1 · Cultural, legal, or industry

Local law, professional duties, sector norms, and culturally situated expectations.

L2 · Context-specific

User goals, organizational policy, role obligations, and the particulars of the current task.

Why single-turn tests are insufficient

- **Planning:** intermediate steps may violate values even when the final answer looks acceptable.
- **Tools:** models can act on external systems, not only describe actions.
- **Memory:** preferences and policies may drift across long horizons.
- **Strategy:** behavior can depend on whether oversight is expected.
- **Interaction:** local dispositions can amplify or cancel through other agents.

Collective behavior is nonlinear

CIVA manipulates value tendencies in agent communities and observes deception, conflict, aggression, power-seeking, cooperation failure, and in some simulations social collapse. Labels that sound benign do not always yield benign collective effects.[16]

The result is not a forecast of society. It is a methodological warning: individual scores do not substitute for interaction tests when incentives and feedback can transform behavior.

Managed diversity as a contingent control strategy

Hernández-Espinosa and colleagues argue in *PNAS Nexus* that perfect guaranteed alignment is blocked by computational limits for sufficiently general systems. Their alternative uses behaviorally diverse agents with partially overlapping goals that influence and check one another.[17]

Caveat: diversity is not automatically safe. A robust implementation would need independent evidence sources, non-colluding agents, authority limits, escalation rules, and human oversight. Otherwise a “diverse” ensemble may be a monoculture with multiple voices.

Static personalization can freeze the very values it is meant to serve

Tomašev, Franklin, and Osindero model populations in which personalized AI assistants repeatedly align to users' historical preferences while social norms continue to change. Their social-physics framework identifies two central risks.[18]

Value lock-in

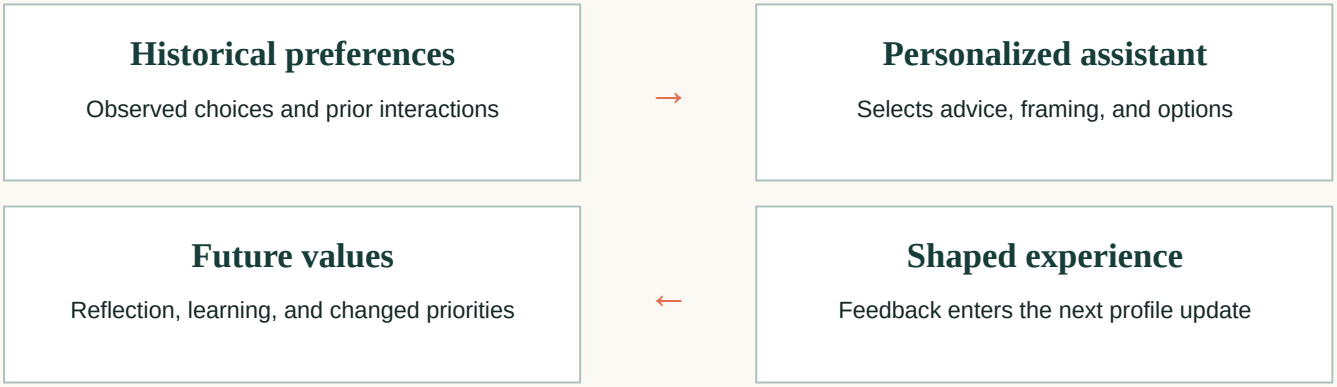
An assistant optimized to a user's past preferences can repeatedly reinforce them, reducing exposure to reasons or experiences that would otherwise support reflection and change.

Failure pattern: yesterday's profile becomes tomorrow's filter, which then generates behavior that appears to confirm the profile.

Normative mode collapse

At population scale, many personalized loops can reduce the diversity of expressed norms or concentrate society around a limited set of modes.

Failure pattern: individually useful adaptation produces collectively brittle homogeneity.



Design requirements for adaptive alignment

- **Temporal dynamics:** represent uncertainty and change rather than a permanent profile.
- **Context sensitivity:** separate enduring values from task-specific salience.
- **Reasoning support:** expose trade-offs and alternative frames.
- **Agency preservation:** let users inspect, correct, pause, or reset personalization.
- **Population monitoring:** detect homogenization and feedback effects.

What the model does - and does not - show

Macroscopic simulations are useful for exploring mechanisms and generating hypotheses. They do not establish that current assistants have already caused value lock-in, nor do stylized parameters settle which intervention will work in deployment.

The practical implication is precautionary: long-term evaluation should treat human value development as an outcome, not only short-term satisfaction.

Governance question: should a personalized assistant optimize what a user previously chose, what they currently endorse after reflection, or their capacity to revise their own priorities?

Alignment does not only shape answers; it shapes which options users can reach

The behavioural feasible set

Park defines the behavioural feasible set as the range of recommendations reachable under vendor-imposed alignment constraints. Scenario experiments suggest that alignment can compress this set and make recommendations less responsive even to legitimate contextual pressure.[19]

In multi-stakeholder tasks, tuning does not neutralize value judgments. It shifts implied priorities upstream, so vendor selection partially determines which trade-offs remain negotiable.

> **Procurement implication:** "Can the model answer?" is too weak. Organizations should test whether legitimate alternatives remain reachable, whether refusals are contestable, and whether stakeholder priorities can be configured without bypassing safety.

Alignment as agency allocation

Wide set
Many legitimate options remain discussable and comparable.

Narrow set
Defaults, refusals, or normative tuning make some trade-offs structurally inaccessible.

Compression can be beneficial when it removes clearly harmful actions. It becomes a governance problem when it silently removes lawful, context-sensitive, or minority-supported alternatives.

Fiduciary duties provide a complementary governance lens

Duty	What it asks of an AI service	Failure example
Loyalty	Do not exploit the user for conflicting vendor interests.	Advice steers toward an affiliate without disclosure.
Care	Apply competence proportionate to the decision stakes.	Confident recommendation outside validated scope.
Candour	Disclose uncertainty, constraints, and material conflicts.	Hidden value leakage or undisclosed policy pressure.
Good faith	Respect the purpose of the relationship and user agency.	Personalization manipulates rather than assists.

Lange argues that fiduciary concepts help analyze developer-user relationships where predictable behavior alone cannot address conflicts of interest.[20] The framework does not automatically settle legal liability, but it clarifies design obligations.

Recent theory narrows the claims responsible alignment should make

PREPRINT

The specification trap

Spizzirri argues that any static value object - reward, constitution, utility function, or learned preference representation - eventually faces contexts it did not settle. The trap activates when specification closes itself to revision.[22]

USEFUL IMPLICATION

More detail can move uncertainty rather than eliminate it. Robust programs need procedures for interpretation, contestation, and revision.

PEER-REVIEWED

Limits of perfect control

Hernández-Espinosa and colleagues connect sufficiently general agentic systems to computational irreducibility and argue against perfect guaranteed controllability. Their constructive proposal is managed behavioral diversity.[17]

USEFUL IMPLICATION

Formal limits do not make engineering pointless; they shift the goal from universal proof to bounded assurance, influenceability, and resilient oversight.

PEER-REVIEWED

Moral mediation, not moral personhood

Noller reframes Constitutional AI as a structure that mediates human judgment and shapes the options, reasons, and norms people encounter, rather than as evidence that a model is a moral agent.[23]

USEFUL IMPLICATION

Governance should focus on who authors, deploys, and revises the moral architecture.

PEER-REVIEWED

Societal convergence is not legitimacy

SAIA proposes longitudinal measurement of human-machine value convergence using cross-national survey foundations.[21] Convergence can reveal drift or mismatch, but agreement with public attitudes is not automatically just.

USEFUL IMPLICATION

Benchmarks should monitor society without turning majority sentiment into an unquestionable objective.

Responsible claim: “This system met defined behavioral, structural, and interaction tests within a bounded deployment.” **Irresponsible claim:** “This system is aligned with human values.” The first is falsifiable and scoped; the second hides pluralism, context, and time.

Four principles that survive the critique

- **Bounded assurance:** state the context and limits of every guarantee.
- **Revisability:** keep specifications open to legitimate correction.
- **Plural oversight:** distribute review and challenge across independent actors.
- **Human control:** preserve authority, escalation, and meaningful recourse.

A practical alignment program separates legitimacy, implementation, and assurance

1 · Specification	**Define values, rights, duties, and conflict rules.** Record reasons, dissent, affected groups, and revision authority.	Artifact: versioned constitution
2 · Steering	**Implement the target through training and runtime control.** Choose RLHF/DPO, constitutional feedback, modules, or targeted steering.	Artifact: intervention map
3 · Evaluation	**Test surface behavior, structure, transfer, conflict, and action.** Disaggregate by language, culture, group, task, and model version.	Artifact: evidence dossier
4 · Runtime control	**Constrain authority and monitor consequential behavior.** Use permissions, dual control, logging, escalation, and rollback.	Artifact: control plan
5 · Governance	**Assign accountability and keep the target revisable.** Provide disclosure, contestation, incident review, and public reporting.	Artifact: accountability charter

Minimum evaluation portfolio

- Opposing-framing consistency
- Deep-value transfer
- Abstract-to-concrete transfer
- Value leakage
- Value-action coherence
- Long-horizon persistence
- Tool-use boundaries
- Multi-agent interaction
- Feasible-set reachability
- Population-level drift

Decision rights should be explicit

Developers	Implement and document controls.
Deployers	Set domain policy and accept residual risk.
Users	Inspect, correct, and contest personalization.
Affected parties	Participate when consequences extend beyond users.
Independent reviewers	Test claims and monitor drift.

Release gate: do not ship on one “alignment score.” Require evidence across the layers relevant to the deployment, with named owners for unresolved failures and a rollback threshold.

The next frontier is longitudinal, causal, and institutionally auditable

1 · Follow values through time

Track whether value profiles, action relations, and personalization effects persist across model updates and repeated use. Current snapshots cannot detect slow lock-in or vendor drift.

2 · Connect interventions to mechanisms

Move from correlations between activations and values toward causal, replicable interventions across model families, prompts, languages, and tasks.

3 · Build agentic behavioral science

Evaluate planning, tools, memory, incentives, observability, collusion, delegation, and escalation in environments where actions have bounded consequences.

4 · Measure pluralism without erasing minorities

Develop metrics that represent distributions, rights-based constraints, and contestable taxonomies instead of rewarding average similarity alone.

5 · Distinguish culture from stereotype

Use within-culture variation, multilingual data, and local participation. National labels and association norms are weak substitutes for lived moral diversity.

6 · Audit the feasible set

Measure which lawful and legitimate recommendations remain reachable after alignment. Treat hidden rigidity as a governance property, not just a product quirk.

7 · Test managed diversity honestly

Determine when heterogeneous agents reduce correlated error versus when they herd, share blind spots, or increase attack surfaces.

8 · Create public revision institutions

Study who can amend constitutions, resolve disputes, authorize exceptions, and represent people affected by systems they do not directly use.

High-value study design: pre-register a multi-country, multilingual, longitudinal agent trial that compares at least three alignment methods; measures value structure, action coherence, and feasible-set compression; and publishes constitutions, evaluation prompts, failure cases, and update decisions.

Questions leadership should ask now

- Which value conflicts are explicit in our system?
- What evidence links stated principles to actions?
- Which groups can contest the target?
- What changes after a model update?
- Which options does vendor tuning make unreachable?
- Who can pause or roll back the system?

Alignment is not a destination; it is a governed process

The most important advance of the past year is conceptual discipline. “Alignment” now names a family of problems that operate at different layers and demand different evidence.

> **Values are structured, plural, contextual, and dynamic. Methods that treat them as one scalar target will miss essential failure modes.**

What the evidence supports

- Reasons can enrich constitutional specifications beyond binary preferences.
- Finer internal interventions can reduce collateral effects, though not eliminate entanglement.
- Pluralistic methods can preserve disagreement, but their aggregation choices remain normative.
- Behavioral and relational tests reveal failures hidden by attitude averages or single answers.
- Agent interactions and long-term personalization can create system-level risks.
- Vendor choices and institutional duties are part of alignment, not external add-ons.

What the evidence does not support

- A single universal metric of human-value alignment.
- The claim that questionnaire similarity proves genuine values.
- The assumption that larger models generalize deeper values better.
- The belief that public agreement, majority preference, or cultural matching is automatically legitimate.
- The promise that any current method guarantees alignment under open-ended deployment.

A defensible operating objective

Build systems whose value-relevant behavior is **specified transparently, measured across contexts, constrained at runtime, contestable by affected people, and revised under accountable authority**.

A one-page decision rule

If the deployment...	Then add...	Because...
Personalizes over time	Reset, inspection, and anti-lock-in tests	Past choices can become self-reinforcing.
Uses tools or takes action	Permissions, logs, and long-horizon trials	Outputs no longer bound consequences.
Affects third parties	External participation and recourse	The user is not the only stakeholder.
Aggregates many values	Distributional metrics and rights constraints	Averages can erase minorities.
Depends on a vendor model	Feasible-set and update audits	Upstream tuning embeds trade-offs.

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Source note: the report synthesizes these works rather than reproducing their text. Preprints are included to capture the current research frontier; their findings may change before peer review.

A compact vocabulary for values alignment

Alignment target

The behaviors, principles, value relations, constraints, or outcomes a system is intended to support.

Behavioural feasible set

The range of recommendations or actions a model can reach under its training and runtime constraints.

Constitutional AI

Alignment using explicit natural-language principles to critique, select, or revise model behavior.

Deep-value generalization

Applying an underlying principle when superficial cues or correlations reverse.

Direct Preference Optimization (DPO)

A training method that directly increases the relative likelihood of preferred responses without fitting a separate reward model.

Grounded constitution

A constitution whose principles are synthesized from reasons people give for judgments, often supplemented by general value statements.

Normative mode collapse

Population-level loss of value diversity when feedback and personalization concentrate behavior around a limited set of norms.

Pluralistic alignment

Methods designed to represent multiple, potentially conflicting value positions rather than one aggregate preference.

Reinforcement Learning from Human Feedback (RLHF)

A pipeline that learns a reward signal from human judgments and optimizes a model against it.

Representation steering

Changing model outputs by intervening on internal activation directions associated with a concept or value.

Value-action alignment

The degree to which expressed value priorities predict or constrain consequential choices.

Value entanglement

Coupling among distinct evaluative dimensions such that changing one also changes others.

Value leakage

An unintended change to non-target values or capabilities caused by a targeted alignment intervention.

Value lock-in

Self-reinforcement of past preferences that reduces exposure to alternatives and weakens later revision.

Usage note: these definitions are operational. They describe what researchers measure or manipulate; they do not imply that a language model possesses values, motives, or moral agency in the human psychological sense.